

BM and GAUSSIAN PROCESSES

Recall on Gaussian r.v.'s

DEF 2.1

A 1-dim. r.v. $\underline{\Gamma}$ is Gaussian if it has the characteristic function

$$\mathbb{E} e^{i \xi \underline{\Gamma}} = e^{i m \xi - \frac{1}{2} \sigma^2 \xi^2} \quad 1)$$

for some real numbers $m \in \mathbb{R}$ and $\sigma > 0$.

RMK 2.2

By differentiating 1) two times it is easy to see that

DEF 2.2

$$m = \mathbb{E} \underline{\Gamma} \quad \sigma^2 = \text{Var} \underline{\Gamma}$$

A random vector $(\underline{\Gamma}_1, \dots, \underline{\Gamma}_n) =: \underline{\Gamma} \in \mathbb{R}^n$ is Gaussian if $\langle \underline{\ell}, \underline{\Gamma} \rangle$ is, for every $\underline{\ell} \in \mathbb{R}^n$ a 1-dim Gaussian.

This is the same requiring that

$$\mathbb{E} e^{i \langle \underline{\xi}, \underline{\Gamma} \rangle} = e^{i \mathbb{E} \langle \underline{\xi}, \underline{\Gamma} \rangle - \frac{1}{2} \text{Var} \langle \underline{\xi}, \underline{\Gamma} \rangle} \quad 2)$$

RMK 2.3

Setting $\underline{m} = (m_1, \dots, m_n) \in \mathbb{R}^n$ and $\Sigma' := (\Sigma'_{jk})_{j,k} = \sigma_{jk}$,

$j, k = 1, \dots, n$ where

$$m_j := m_j \quad \text{and} \quad \sigma_{jk} := \mathbb{E} (\underline{\Gamma}_j - m_j) (\underline{\Gamma}_k - m_k) = \text{Cov} (\underline{\Gamma}_j, \underline{\Gamma}_k)$$

$$\text{Var} \langle \underline{\zeta}, \underline{\Sigma} \rangle = \text{Var} \langle \underline{\zeta}, \underline{\Sigma} \rangle + \text{Var} \langle \underline{\zeta}, \underline{\Sigma} \rangle + \dots + \text{Var} \langle \underline{\zeta}, \underline{\Sigma} \rangle + \text{all Gaussian}$$

it possible to rewrite 2) as

DEF 2.4

$$\mathbb{E} e^{i \langle \underline{\zeta}, \underline{\Sigma} \rangle} = e^{i \langle \underline{\zeta}, \underline{m} \rangle - \frac{1}{2} \langle \underline{\zeta}, \underline{\Sigma} \underline{\zeta} \rangle}$$

The vector $\underline{m}$ is called the mean vector,

the matrix $\underline{\Sigma}$ is called the variance and covariance matrix.

The first obtainable properties of BM follow from the definition which are related with the properties of the Gaussian r.v.s.

PROP 2.5 — 1-dim BM

It follows by definition that $\mathbb{E} e^{i \underline{\zeta} B_t} = e^{-\frac{t}{2} \underline{\zeta}^2}$ for all $t \geq 0$, $\underline{\zeta} \in \mathbb{R}$. It is possible to see that the integral $\mathbb{E} e^{i \underline{\zeta} B_t}$ is convergent for any $\underline{\zeta} \in \mathbb{C}$. Indeed, take $\text{Re} \underline{\zeta} = 0$, then

$$\begin{aligned} & \int_{-\infty}^{+\infty} e^{i \underline{\zeta} x} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx = \int_{-\infty}^{+\infty} e^{i \text{Im}(\underline{\zeta}) x} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx \\ & = \int_{-\infty}^{+\infty} e^{x^2} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx = \int_{-\infty}^{+\infty} e^{x^2} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx < +\infty \end{aligned}$$

where we used that

$$e^{(y_1 - c)^2} \geq 1 \rightarrow e^{cy} \leq e^{c^2 + y^2/4} \rightarrow e^{cy} e^{-y^2/2} \leq e^{c^2} e^{-y^2/4}.$$

It follows that

$$\mathbb{E} e^{z B_t} = e^{t z^2/2} \text{ for all } z \in \mathbb{C}.$$

PROP 2.6

Moments of BM can be obtained with a direct computing.

Indeed, for $k = 0, 1, 2, \dots$

$$\mathbb{E} B_t^{2k+1} = \underbrace{\int_{-\infty}^{+\infty} x^{2k+1} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx}_{\text{odd}} = 0$$

$$\begin{aligned} \mathbb{E} B_t^{2k} &= \int_{-\infty}^{+\infty} x^{2k} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx \\ &= 2 \int_0^{\infty} x^{2k} \frac{e^{-\frac{x^2}{2t}}}{\sqrt{2\pi t}} dx \\ &= 2 \int_0^{\infty} (2ty)^k \frac{e^{-y}}{\sqrt{2\pi t}} \sqrt{2t} \frac{1}{2} y^{-1/2} dy \quad \left\{ \begin{array}{l} \frac{x^2}{2t} = y \\ x = \sqrt{2ty} \\ dx = \sqrt{2t} \frac{1}{2} y^{-1/2} dy \end{array} \right. \\ &= 2^k t^k \frac{1}{\sqrt{\pi}} \int_0^{\infty} y^{k-1/2} e^{-y} dy \\ &= \frac{2^k t^k}{\sqrt{\pi}} \int_0^{\infty} y^{k+1/2-1} e^{-y} dy \end{aligned}$$

$$= \frac{2^n t^{n/2}}{\sqrt{\pi}} \triangleq (n+1/2).$$

It follows in particular that

COR 2.7

$$\mathbb{E} B_t = \mathbb{E} B_t^3 = \mathbb{E} B_t^5 = \dots = 0$$

$$\text{Var } B_t = \mathbb{E} B_t^2 = t \quad \mathbb{E} B_t^4 = 3t^2.$$

$$\text{Cov}(B_s, B_t) = \mathbb{E} B_s B_t = s \wedge t$$

↳ to see this, let $s \leq t$,

$$\mathbb{E} B_s B_t = \mathbb{E} (B_s (B_t - B_s) + B_s^2)$$

indep
of increment.

$$= \underbrace{\mathbb{E} B_s}_{=0} \underbrace{\mathbb{E} (B_t - B_s)}_{=0} + \underbrace{\mathbb{E} B_s^2}_s = s.$$

DEF 2.8

Definition. A 1-dim stochastic process $(X_t)_{t \geq 0}$ is called a Gaussian process if all vectors

$$(X_{t_1}, \dots, X_{t_n}), \quad n \geq 1, \quad 0 \leq t_1 < t_2 < \dots < t_n$$

are Gaussian (possibly degenerate).

Theorem. A 1-dim BM is a Gaussian process. Further for $t_0 := 0 < t_1 < \dots < t_n$, $n \geq 1$, the vector

$$(B_{t_1}, \dots, B_{t_n})^T =: \underline{\Gamma}$$

is a Gaussian r.v. with variance covariance matrix C :

$(C)_{jk} = (t_j \wedge t_k)_{j,k=1,\dots,n}$ and $\underline{m} = 0$. Moreover it admits the density function

$$\frac{1}{(2\pi)^{n/2}} \frac{1}{\sqrt{\det C}} e^{-\frac{1}{2} \langle \underline{x}, C^{-1} \underline{x} \rangle} = \frac{1}{(2\pi)^{n/2} \sqrt{\prod_{s=1}^n (t_s - t_{s-1})}} e^{-\frac{1}{2} \sum_{s=1}^n \frac{(x_s - x_{s-1})^2}{t_s - t_{s-1}}}$$

Proof. We should prove that

$$\mathbb{P} e^{\langle \underline{x}, \underline{\Gamma} \rangle} = e^{-\frac{1}{2} \langle \underline{x}, C \underline{x} \rangle}$$

Observe that

$$\underline{\Gamma} = \begin{pmatrix} 0 & \dots & 0 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{pmatrix} \stackrel{=: M}{=} \begin{pmatrix} B_{t_1} - B_{t_0} \\ \vdots \\ B_{t_n} - B_{t_{n-1}} \end{pmatrix} \stackrel{=: \Delta}{=} \begin{pmatrix} B_{t_1} - B_{t_0} \\ \vdots \\ B_{t_n} - B_{t_{n-1}} \end{pmatrix}$$

$n \times n$ lower triangular $n \times 1$

Since

$$B_{t_k} - B_{t_0} = \sum_{s=1}^k (B_{t_s} - B_{t_{s-1}}) \quad \left(\begin{matrix} \text{well} \\ t_0 = 0 \end{matrix} \right)$$

Hence

$$\begin{aligned}
 E e^{i \sum_{j=1}^m \xi_j} &= E e^{i \langle \xi, M \Delta \rangle} \\
 &= E e^{i \langle M^T \xi, \Delta \rangle} \quad M^T \Delta = \begin{pmatrix} \xi_1 + \dots + \xi_m \\ \xi_2 + \dots + \xi_m \\ \vdots \\ \xi_m \end{pmatrix} \\
 &= E e^{i \sum_{j=1}^m (B_{t_j} - B_{t_{j-1}}) (\xi_1 + \dots + \xi_m)}
 \end{aligned}$$

indep. of increments
Gaussianity

$$\begin{aligned}
 &= \frac{1}{\pi} E e^{i \sum_{j=1}^m (B_{t_j} - B_{t_{j-1}}) (\xi_1 + \dots + \xi_m)} \\
 &= \frac{1}{\pi} E e^{-\frac{1}{2} \sum_{j=1}^m (t_j - t_{j-1}) (\xi_1 + \dots + \xi_m)^2} \\
 &= e^{-\frac{1}{2} \left[\sum_{j=1}^m t_j (\xi_1 + \dots + \xi_m)^2 - \sum_{j=1}^{m-1} t_{j-1} (\xi_1 + \dots + \xi_m)^2 \right]}
 \end{aligned}$$

Note that

$$\begin{aligned}
 &\sum_{j=1}^m t_j (\xi_1 + \dots + \xi_m)^2 - \sum_{j=1}^{m-1} t_{j-1} (\xi_1 + \dots + \xi_m)^2 \\
 &\quad \text{last term} \\
 &= t_m \xi_m^2 + \sum_{j=1}^{m-1} t_j (\xi_1 + \dots + \xi_m)^2 - \sum_{j'=0}^{m-1} t_{j'} (\xi_{j'+1} + \dots + \xi_m)^2
 \end{aligned}$$

↳ the term $j'=0$ is zero since $t_{j'}=0$

$$= t_m \xi_m^2 + \sum_{j=1}^{m-1} t_j \left[(\xi_1 + \dots + \xi_m)^2 - (\xi_{j+1} + \dots + \xi_m)^2 \right]$$

$$\begin{aligned}
 &\xi_1^2 + \dots + \xi_m^2 + \sum_{\substack{k \neq l \\ 1 \leq k, l \leq m}} \xi_k \xi_l - \xi_{j+1}^2 - \dots - \xi_m^2 - \sum_{\substack{k \neq l \\ j+1 \leq k, l \leq m}} \xi_k \xi_l \\
 &= \xi_j^2 + 2 \sum_{k=j+1}^m \xi_j \xi_k = \xi_j \left(\xi_j + 2 \sum_{k=j+1}^m \xi_k \right)
 \end{aligned}$$

$$= t_m \xi_m^2 + \sum_{j=1}^{m-1} t_j \xi_j \left(\xi_j + 2 \xi_{j+1} + \dots + 2 \xi_m \right)$$

$$= t_m \xi_m^2 + t_1 \xi_1^2 + 2 t_1 \xi_1 \xi_2 + \dots + 2 t_1 \xi_1 \xi_m$$

$$+ \dots + t_{n-1} \xi_{n-1}^2 + t_{n-1} \xi_{n-1} \xi_n$$

$$= \sum_{j=1}^n \sum_{k=1}^n (t_j + t_k) \xi_j \xi_k$$

Hence $\sigma e^{i\langle x, \xi \rangle} = e^{-1/2 \langle \xi, C \xi \rangle}$

where

$$C = \begin{bmatrix} t_1 & t_1 & \dots & t_1 \\ t_1 & t_1 + t_2 & \dots & t_2 \\ \vdots & t_2 & \ddots & \vdots \\ t_1 & t_2 & \dots & t_n \end{bmatrix}$$

$\hookrightarrow$ Strictly positive definite and symmetric

$\hookrightarrow$ inverse C^{-1} is also positive definite.

In order to see that μ admits the density above, it is possible to directly compute

$$\int_{\mathbb{R}} e^{i\langle x, \xi \rangle} (2\pi)^{-\frac{n}{2}} (\det C)^{-1/2} e^{-1/2 \langle x, C^{-1} x \rangle} dx$$

$$= e^{-1/2 \langle \xi, C \xi \rangle}$$

and it can be directly computed that

$$\langle x, C^{-1} x \rangle = \sum_{s=1}^n \frac{(x_s - x_{s-1})^2}{t_s - t_{s-1}}$$

Inspection of the proof above actually shows more than the fact the BM is a Gaussian process.

THM 2.10

Theorem. Let $(X_t)_{t \geq 0}$ be a 1-dim Gaussian process with $\underline{\Gamma} := (X_{t_i} - X_{t_{i-1}})^\top$ is Gaussian with mean 0 and covariance $C := (C)_{s,n} = (t_s \wedge t_n)_{s,n=1,\dots,m}$. If $(X_t)_{t \geq 0}$ has continuous sample paths, then $(X_t)_{t \geq 0}$ is a 1-dim BM.

Proof Continuity of paths is assumed. Then $B_0 \sim N(0,0) = \delta_0$. Set $\Delta = (X_{t_1} - X_{t_0}, \dots, X_{t_m} - X_{t_{m-1}})$ and let M be or above, so that, or above

$$\underline{\Gamma} = M \Delta \quad \rightarrow \quad \Delta = M^{-1} \underline{\Gamma}$$

We directly compute

$$\mathbb{P} e^{i \langle \xi, \Delta \rangle} = \mathbb{P} e^{i \langle \xi, M^{-1} \underline{\Gamma} \rangle} = \mathbb{P} e^{i \langle (M^{-1})^\top \xi, \underline{\Gamma} \rangle}$$

$$\begin{aligned} \stackrel{\text{Gaussian}}{=} & e^{-\frac{1}{2} \langle (M^{-1})^\top \xi, C (M^{-1})^\top \xi \rangle} \\ = & e^{-\frac{1}{2} \langle \xi, M^{-1} C (M^{-1})^\top \xi \rangle} \end{aligned}$$

One can see that

$$M^{-1} \subseteq (U^{-1})^T = \begin{pmatrix} t_1 - t_0 & & & \\ & t_2 - t_1 & & \\ & & \ddots & \\ & & & t_n - t_{n-1} \end{pmatrix}$$

Thus Δ is Gaussian with uncorrelated (and thus independent), components with $N(0, t_j - t_{j-1})$ distrib.
 Thus prove that increments are indep, stationary (and Gaussian).

INVARIANT PROPERTIES

Various transformations of a BM are still BM.

For example:

PROP 2.11

- Reflection of BM. If $(B_t)_{t \geq 0}$ is a BM then $(X_t)_{t \geq 0}$ where $X_t := -B_t$ is still a BM.

Indeed

0) $X_0 = -B_0 = 0$ a.s.

1) Take $t_0 \leq t_1 \leq \dots \leq t_3 \leq \dots \leq t_n$, we have
 $X_{t_1} - X_{t_{j-1}} = -B_{t_3} + B_{t_{j-1}} = -(B_{t_3} - B_{t_{j-1}})$ and
 this indep. follows by indep of $B_{t_3} - B_{t_{j-1}}$

$$\begin{aligned}
 2) \text{ For } s \leq t, \quad X_{t+h} - X_{s+h} &= - (B_{t+h} - B_{s+h}) \\
 &\sim - (B_t - B_s) \\
 &= X_t - X_s
 \end{aligned}$$

for any $h \geq 0$

$$\begin{aligned}
 3) \quad X_t - X_s &= - (B_t - B_s) \sim N(0, t-s) \\
 &= N(0, t-s)
 \end{aligned}$$

4) Gaussianity follows trivially.

- Remark. Let B_t be a BM and fix $a \geq 0$ arbitrary. Then $(W_t)_{t \geq 0}$, $W_t := B(t+a) - B(a)$ is again a BM.

$$0) \quad W_0 = B(a) - B(a) = 0$$

$$\begin{aligned}
 1) \quad W_{t_j} - W_{t_{j-1}} &= B(t_j + a) - B(a) - B(t_{j-1} + a) + B(a) \\
 &= B(t_j + a) - B(t_{j-1} + a)
 \end{aligned}$$

$$\begin{aligned}
 2) \quad W_{t+h} - W_{s+h} &= \underbrace{B_{t+h}^a - B_{s+h}^a}_{\text{ indep. since } B \text{ is BM}} \sim B_t - B_s \\
 &= W_t - W_s
 \end{aligned}$$

$$3) \quad W_t - W_s = B_t^a - B_s^a \sim N(0, t-s)$$

4) trivial.

RMK 2.12

A consequence of independence of increments is that BM has no memory, in the sense that it has the Markov property.

LEMMA 2.13

Lemma. Let $(B_t)_{t \geq 0}$ be a BM. Let $W_t := B_{t+a} - B_a$. Then $(B_t)_{0 \leq t \leq a}$ and $(W_t)_{t \geq 0}$ are indep., i.e., the σ -Algebras generated by them are indep.:

$$\sigma(B_t : 0 \leq t \leq a) =: \mathcal{F}_{a, B}^B \perp \mathcal{F}_a^W := \sigma(W_t : 0 \leq t < +\infty)$$

In particular $B_t - B_r \perp \mathcal{F}_r^B$ for all $r \leq t$.

Proof. First note that, for arbitrary r.v.'s $X_0, X_1, \dots, X_m$, it is true that

$$(*) \quad \sigma(X_j : j=0 \dots m) = \sigma(X_0, X_j - X_{j-1} : j=1 \dots m).$$

To see this note that X_0 and $X_j - X_{j-1}$, $j=1 \dots m$, are independent wrt $\sigma(X_j : j=0 \dots m)$, this proves the inclusion $\supset$. The reverse inclusion instead follows

since $X_k = \left[\sum_{j=1}^k (X_j - X_{j-1}) \right] + X_0$, $k=1 \dots m$, and

thus X_k , $k=1 \dots m$, are $\sigma(X_0, X_j - X_{j-1} : j=1 \dots m)$ measurable. Hence the inclusion $\subset$ follows.

Take now

$$0 = r_0 < r_1 < \dots < r_m \leq a \quad t_0 < t_1 < \dots < t_n.$$

We prove now that the class of events

$$\bigcup_{m \in \mathbb{N}} \bigcup_{r_0 < r_1 < \dots < r_m \leq a} \sigma(B_{r_j} : j=1, \dots, m) \perp\!\!\!\perp \bigcup_{m \in \mathbb{N}} \bigcup_{0 \leq t_1 < t_2 < \dots < t_n \leq a} \sigma(W_{t_j} : j=1, \dots, n) \quad \textcircled{*}$$

⌊ Needed for the law being a σ -system

are independent. Since they are σ -systems it follows that the generated σ -Algebras are indep., i.e., $\mathcal{F}_a^B$ and $\mathcal{F}_a^W$ are indep.

The fact that $\textcircled{*}$ is true follows since

$B_{r_1} - B_{r_0}, \dots, B_{r_m} - B_{r_{m-1}}, B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}$ are indep. by indep of increments and we can use $\textcircled{*}$ to get

$$\sigma(B_{r_j}, j=1, \dots, m) \perp\!\!\!\perp \sigma(\underbrace{B_{t_k} - B_{t_0}}_{\mathcal{F}_a^W}, k=1, \dots, n).$$

Since $r_j, j=1, \dots, m, t_k, k=1, \dots, n$ and a are all arbitrary, the result follows.

Finally, taking $a=r$, we see that

$$W(t-r) = B(t) - B(r)$$

which $\mathcal{F}_r^W$ -measurable. Hence $B(t) - B(r) \perp \mathcal{F}_r^B$.

Another useful property of BM is time inversion.

— Time inversion. Let $(B_t)_{t \geq 0}$ be a BM and fix some time $a > 0$. Then $W_t := B_{a-t} - B_a$, $t \in [0, a]$ is again a BM. This follows very much from the renewal property. Indeed

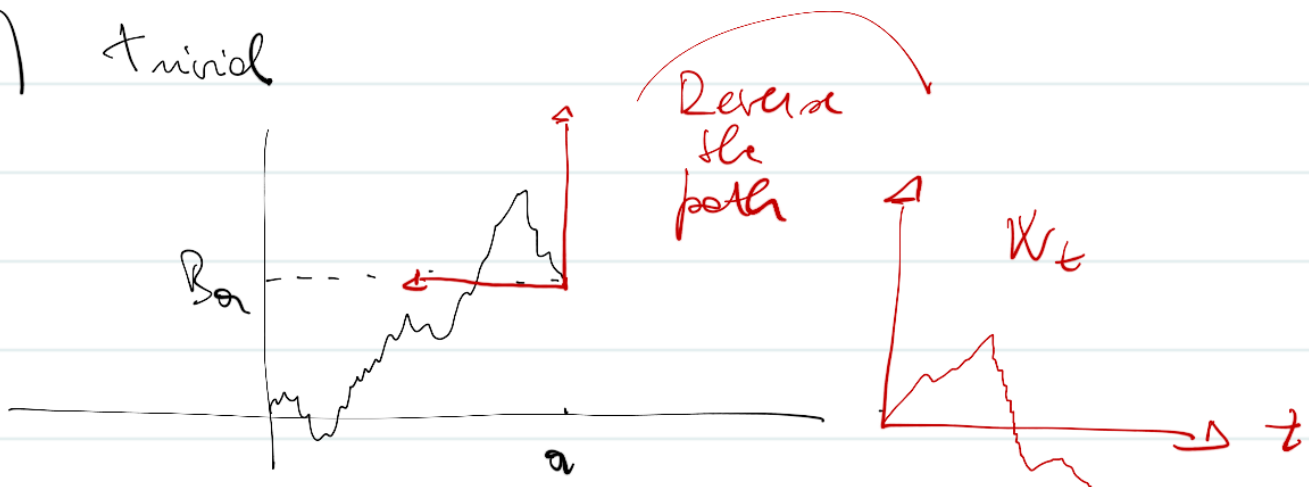
$$0) W_0 = B_a - B_a = 0$$

$$\begin{aligned} 1) W_{t_s} - W_{t_{s-1}} &= B_{a-t_s} - B_a - B_{a-t_{s-1}} + B_a \\ &= - \underbrace{(B_{a-t_{s-1}} - B_{a-t_s})}_{\text{indep. of } s=1, \dots, n} \end{aligned}$$

$$2) W_{t_n} - W_{t_{n-1}} = - (B_{a-t_{n-1}} - B_{a-t_n}) \sim (B_{a-t_n} - B_{a-t_{n-1}})$$

$$\begin{aligned} 3) W_t - W_r &= - (B_{a-r} - B_{a-t}) \sim -N(0, t-r) \\ &= N(0, t-r) \end{aligned}$$

4) $\uparrow$ trivial



Here is stated the well known scaling property of BM.

Scaling. For all $c > 0$ and $t \geq 0$, we have that

$$B_{ct} \sim \sqrt{c} B_t$$

In particular $(c^{-1/2} B_{ct})_{t \geq 0}$ is again a BM.

↳ A different scaling does not change indep. of increments or continuity. Further $N(0, ct) = \sqrt{c} N(0, t)$

DEF 2.15

— BM in $\mathbb{R}^d$

The goal now is to make clear that a process in $\mathbb{R}^d$ $B_t = (B_t^1 \dots B_t^d)$ is a BM if and only if the coordinate processes B_t^i , $i=1, \dots, d$ are indep.

1-dim. BMs. Independence of processes is meant in the sense that two processes $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ are indep. if

$$\mathcal{F}_\infty^X \perp \mathcal{F}_\infty^Y$$

where

$$\mathcal{F}_\infty^X := \sigma \left(\bigcup_{n \geq 1} \bigcup_{\substack{0 \leq t_1 < \dots < t_n < \infty}} \sigma(X_{t_1}, \dots, X_{t_n}) \right)$$

Note that, as we already explained before, the σ -algebra $\mathcal{F}_0^X$ is generated by a σ -system. Hence indep. of the σ -Algebra follows by indep. of the σ -system. Hence we should prove

$$(X_{s_1}, \dots, X_{s_m}) \perp (Y_{t_1}, \dots, Y_{t_m})$$

for every $m, n \geq 1$, $s_1 < \dots < s_m$ and $t_1 < \dots < t_m$.

Indeed suppose that we have

$$(X_{s_1}, X_{s_2}) \perp (Y_{t_1}, Y_{t_2}) \text{ for all } s_1, s_2 > 0. \quad (*)$$

It is true that

$$X_s \perp Y_t \text{ for } s, t > 0 \text{ arbitrarily?}$$

$$\begin{aligned} P(X_s \in B_1, Y_t \in B_2) &= \\ &= P(X_s \in B_1, X_t \in \mathbb{R}, Y_s \in \mathbb{R}, Y_t \in B_2) \end{aligned}$$

$$\begin{aligned} \text{by } (*) &= P(X_s \in B_1, X_t \in \mathbb{R}) P(Y_s \in \mathbb{R}, Y_t \in B_2) \\ &= P(X_s \in B_1) P(Y_t \in B_2) \end{aligned}$$

By generalizing this idea the result follows.

It will be useful to use the following characterization of BM.

LEMMA 2.17

Lemma. Let $(X_t)_{t \geq 0}$ be a d -dim stochastic process. Then it satisfies 0) - 3) iff, for all $n \geq 0$, and

$0 = t_0 < t_1 < \dots < t_n$, and $\xi_0 \dots \xi_n \in \mathbb{R}^d$, $\mathbb{E} e^{i \langle \xi, X \rangle} = \exp \left[\sum_{k=1}^n \langle \xi_k, X_{t_k} - X_{t_{k-1}} \rangle \right]$

$$\begin{aligned}
 & \mathbb{P} \left[\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right] \\
 &= \exp \left(-\frac{1}{2} \sum_{j=1}^n |\xi_j|^2 (t_j - t_{j-1}) \right). \quad (*)
 \end{aligned}$$

Hence if X has continuous sample paths then it is a BM^d.

Proof Assume that X satisfies 0) ... 3). Since the characteristic function of a Gaussian r.v., with covariance matrix $(t-s)Id$ and 0 expectation, is $\exp(-\frac{1}{2}(t-s)|\xi|^2)$, we get

$$\begin{aligned}
 & \mathbb{P} \left[\exp \left(i \sum_{j=1}^n \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle + i \langle \xi_0, X_{t_0} \rangle \right) \right] \\
 &= 1) \left(\prod_{j=1}^n \mathbb{P} \left[e^{i \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle} \right] \right) \underbrace{\mathbb{P} \left[e^{i \langle \xi_0, X_{t_0} \rangle} \right]}_{= 1 \text{ by 0)}
 \end{aligned}$$

$$2) \quad \prod_{j=1}^n \mathbb{P} \left[\exp \left(i \langle \xi_j, X_{t_j} - X_{t_{j-1}} \rangle \right) \right]$$

$$3) \quad \prod_{j=1}^n \exp \left(-\frac{1}{2} (t_j - t_{j-1}) |\xi_j|^2 \right)$$

Conversely, assume that the statement $\textcircled{A}$ holds.
 Fix $k \in \{0, \dots, n\}$ and pick in $\textcircled{A}$ $\xi_j = 0$ for every
 $j \neq k$. Then $\textcircled{A}$ reads

$$\mathbb{E} e^{i \tau \xi_k, X_{t_n} - X_{t_{n-1}}} = e^{-\frac{1}{2} |\xi_k|^2 (t_n - t_{n-1})}$$

from which 3) follows.

But also

$$\mathbb{E} e^{i \tau \xi_k, X_{t_n} - X_{t_{n-1}}} = e^{-\frac{1}{2} |\xi_k|^2 (t_n - t_{n-1})}$$

Hence

$$\mathbb{E} e^{i \tau \xi_k, X_{t_k} - X_{t_{k-1}}} = \mathbb{E} e^{i \tau \xi_k, X_{t_n} - X_{t_{n-1}}}$$

from which 2) follows.

We also have that 1)

is implied by $\textcircled{A}$. Then by letting $n \rightarrow \infty$ 0) follows.)

Since $\textcircled{A}$ is the factorization of the char functions
 of $X_{t_n} - X_{t_{n-1}}$ (Kac theorem)

RMK 2.18

It follows from previous result that if B_t is
 B_n^d then the coordinate processes are 1-dim,
 indep. BM's.

COR 2.19

Corollary. Let B be a B_n^d . Then the coordinate

processes B^j , $j=1, \dots, d$, are indep - BM¹. J

Proof. Since B is BM^d then it satisfies $\star$.
 Use $\star$ for $\sum_{j=1}^d z_j e_k$ where $e_k = [0 \dots \underbrace{1 \dots 0}_d]^T$

for all $j=1, \dots, n$ and any arbitrary fixed $n \in \mathbb{N}$.
 Then B^k satisfies $\star$ for all $z_1, \dots, z_n \in \mathbb{R}$.

Continuity of paths of B^j follows from continuity of paths of B . Hence B^j is a BM¹ for every $j=1, \dots, d$. Now we should prove that these 1-dim BMs are indep. Hence we should verify that $(F_t^{B^1})_{t \geq 0}, (F_t^{B^2})_{t \geq 0}$ are indep., i.e., that

$(B_{t_1}^k, \dots, B_{t_n}^k)$ $k=1, 2, \dots, d$ are indep.

for any choice of $n \in \mathbb{N}$ and $t_0=0 \leq t_1 \leq \dots \leq t_n$.

Since each $B_{t_i}^k$, $k=1, \dots, n$, can be written as

$B_{t_i}^k = \sum_{j=1}^i (B_{t_j}^k - B_{t_{j-1}}^k)$ it is enough to show that all the increments $(B_{t_j}^k - B_{t_{j-1}}^k)$, $j=1, \dots, n$, $k=1, \dots, d$ are indep. This follows again from $\star$ since

Let $\sum_{k=1}^d B^k = \left(\sum_{k=1}^d B^k, \dots, \sum_{k=1}^d B^k \right)$

$$E e^{i \sum_{j=1}^n \langle \xi_j^k, B \rangle}$$

$$B^n = (B_{t_1}^k - B_{t_0}^k, \dots, B_{t_n}^k - B_{t_{n-1}}^k)$$

$$E \exp \left(i \sum_{j=1}^n \sum_{k=1}^d \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k) \right) = E \exp \left(i \sum_{j=1}^n \langle \xi_j, B_{t_j} - B_{t_{j-1}} \rangle \right)$$

$$= \exp \left(-\frac{1}{2} \sum_{j=1}^n |\xi_j|^2 (t_j - t_{j-1}) \right)$$

$$= \exp \left(-\frac{1}{2} \sum_{j=1}^n \sum_{k=1}^d (\xi_j^k)^2 (t_j - t_{j-1}) \right)$$

$$= \prod_{j=1}^n \prod_{k=1}^d \exp \left(-\frac{1}{2} (\xi_j^k)^2 (t_j - t_{j-1}) \right)$$

$$= \prod_{j=1}^n \prod_{k=1}^d E e^{i \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k)} = \prod_{k=1}^d E e^{i \langle \xi, B \rangle^k}$$

Since the B_t^k are BM's, Independence This follows.

Since indep. for fixed k one index or

The converse of this equality is also true. Hence we can prove the following theorem, for fixed k for BM

THM 2.29

Theorem. B is a BM^d iff the coordinate processes $B^1, \dots, B^d$ are indep. BM^1 .

Proof. By previous equality we should prove only the if part. Hence we check that B is a BM^d

if B_t^j are BM^k for $j=1 \dots d$.

then fix n , $t_0=0 < t_1 < \dots < t_n < +\infty$, $\xi_j^k \in \mathbb{R}$, $j=1 \dots n$, $k=1 \dots d$. By assumption, each B^j satisfies Δ , i.e., for all $k=1 \dots d$

$$\begin{aligned} & \mathbb{E} \left[\exp \left(i \sum_{j=1}^n \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k) \right) \right] \\ &= \exp \left(-\frac{1}{2} \sum_{j=1}^n (\xi_j^k)^2 (t_j - t_{j-1}) \right). \end{aligned}$$

Multiplying both sides of the equalities above for $k=1 \dots d$ and using indep. of B^j , $j=1 \dots k$

we get

$$\begin{aligned} & \mathbb{E} \left[\prod_{k=1}^d \exp \left(i \sum_{j=1}^n \xi_j^k (B_{t_j}^k - B_{t_{j-1}}^k) \right) \right] \\ &= \exp \left(-\frac{1}{2} \sum_{k=1}^d \sum_{j=1}^n (\xi_j^k)^2 (t_j - t_{j-1}) \right) \end{aligned}$$

which is exactly Δ for $B = (B^1 \dots B^d)$
and $\xi_j = (\xi_j^1 \dots \xi_j^d) \in \mathbb{R}^d$. The result follows. $\square$